

SHETH MVLU COLLEGE

SUBJECT:- Tableau

Practical 9

AIM:-

- a) Identify and handle missing or inconsistent data
- b) Convert wide data into tall format
- c) Perform union of multiple files
- d) Execute cross-database joins
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets
- g) Apply advanced data cleaning techniques

Dataset: Custom Designed Students Dataset [csv files]

You can create different csv files using Notepad/text Editor. Copy paste the data in Notepad/text Editor and save the files with .csv extension

1) Sem1_Scores.csv

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	90%	88	95	85 %
S102	priya VERMA	ECE-02	75%	80	85	90	%
S103	Rohan Gupta	CSE-01	78	82%	88	78	%
S104	Sneha Patel	IT-03	92	88%	90	96	95 %
S105	aditya Rao	MECH-04	45	50	60	65	%
S106	Ananya Singh	CSE-01	88	91%	85	92	88 %
S107	Rahul Nair	ECE-02	70%	75	80	72	%
S108	Kavya Reddy	IT-03	95	94%	92	98	98 %
S109	Vikram Joshi	CSE-01	67	60%	58	72	80 %
S110	Pooja Iyer	ECE-02	82	85%	80	89	91 %

2) Sem2_Scores. Csv

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	88	92%	90	98	88 %
S102	Priya Verma	ECE-02	79	78%	82	88	92 %
S103	Rohan Gupta	CSE-01	82	85%	80	90	80 %
S104	Sneha Patel	IT-03	95	90%	94	98	96 %
S105	Aditya Rao	MECH-04	52	58%	55	65	70 %
S106	Ananya Singh	CSE-01	90	94%	88	95	90 %
S107	Rahul Nair	ECE-02	74	73%	78	82	75 %
S108	Kavya Reddy	IT-03	96	96%	95	99	99 %

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S109,Vikram Joshi,CSE-01,70,65%,62,75,82 %

S110,Pooja Iyer,ECE-02,85,88%,84,92,94 %

3) Department_Master.csv

Dept_Code,Department_Name,HOD_Name,Dept_Target_Average>Total_Dept_Capacity

CSE-01,Computer Science,Dr. Raman,85,120

ECE-02,Electronics & Comm,Dr. Iyer,78,90

IT-03,Information Tech,Dr. Sen,82,60

MECH-04,Mechanical Engg,Dr. Sharma,70,60

4) Student_Activities.csv

Student_ID,Club_Membership,Projects_Completed,Certifications

S101,Coding Club,3,2

S102,Robotics Club,2,1

S103,Coding Club,2,1

S104,Debate Society,4,3

S105,Sports Club,1,0

S106,Coding Club,3,2

S107,Robotics Club,2,1

S108,Debate Society,4,3

S109,Sports Club,1,1

S110,Cultural Club,2,2

A: Identify and Handle Missing or Inconsistent Data

Objective: Detect and resolve data quality issues including missing records (Null values) and data type mismatches (numerical values formatted as text strings with percentage signs).

Tableau treats fields containing non-numeric characters (like % or extra whitespace) as Text (Abc), which disables numerical operations such as calculating averages, standard deviations, or plotting continuous measures. Missing exam entries (Null) cause dropped rows or incorrect aggregations unless properly imputed or handled.

Step 1: Load the Dataset

1. Open Tableau Desktop.
2. Under the Connect pane on the left, click Text file.
3. Select and open Sem1_Scores.csv.
4. In the Data Source preview table: o Observe missing entries (Null) in Math, Physics, and Chemistry.

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

deliveries.csv

Department_Master.csv

matches.csv

modified_data.csv

Sem1_Scores.csv

Sem2_Scores.Csv.txt

store_sales_data (2).csv

Student_Activities.csv

zomato.csv

New Union

New Table Extension

Sem1_Scores.csv

8 fields 10 rows

Name Sem1_Scores.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Abc	Student ID	Sem1_Scores.csv	Student_ID
Abc	Full Name	Sem1_Scores.csv	Full_Name
Abc	Dept Info	Sem1_Scores.csv	Dept_info

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85
S102	priya VERMA	ECE-02	null	0.750000	80	85	90
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	78
S104	Sneha Patel	IT-03	92	0.880000	90	96	95
S105	aditya Rao	MECH-04	45	null	50	60	65
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88

Go to Worksheet

Data Source

Sheet 1

29°

Search

ENG IN

70%

20:25

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Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)

1. Navigate to Sheet 1 (bottom-left tab).
2. Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field....

3. Create the cleaned measure for Physics:

- Field Name: Physics_Clean
- Tableau Formula: Code snippet

`FLOAT(REPLACE(STR([Physics]), '%', ''))`

- Click OK.

4. Create the cleaned measure for Attendance:

- Right-click the Data Pane --> Create Calculated Field....

- Field Name: Attendance_Clean

- Tableau Formula: Code snippet

`FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`

- Click OK.

Step 3: Handle Missing (Null) Numeric Values via Imputation

1. Handle missing entries in Math:

- Right-click the Data Pane --> Create Calculated Field....

- Field Name: Math_Clean

- Tableau Formula: Code snippet

`IFNULL([Math], 0)`

- Click OK.

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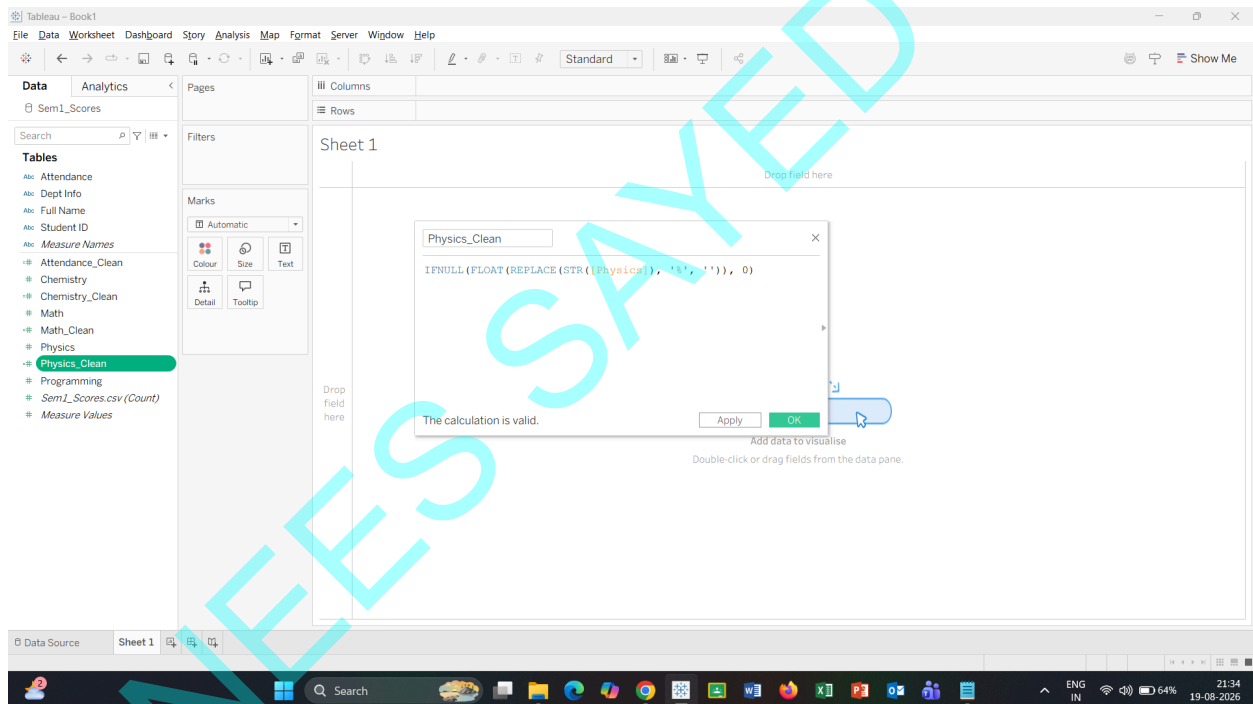
2. Handle missing entries in Chemistry:

- Right-click the Data Pane --> Create Calculated Field....
- Field Name: Chemistry_Clean
- Tableau Formula: Code snippet
IFNULL([Chemistry], 0)
- Click OK.

Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

1. Student S105 contains a Null value in Physics. Combine string stripping with null handling:

- Edit Physics_Clean (right-click Physics_Clean --> Edit...):
- Updated Tableau Formula: Code snippet
IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)
- Click OK.



B: Convert Wide Data into Tall Format

Objective: Transform the horizontal subject columns (Math, Physics, Chemistry, Programming) into a normalized, vertical key-value pair: Subject (Dimension) and Score (Measure).

In wide format, comparing performance across subjects requires manually adding multiple distinct measures to shelves. Unpivoting to tall (tidy) format collapses four separate columns into a single categorical dimension (Subject), enabling dynamic

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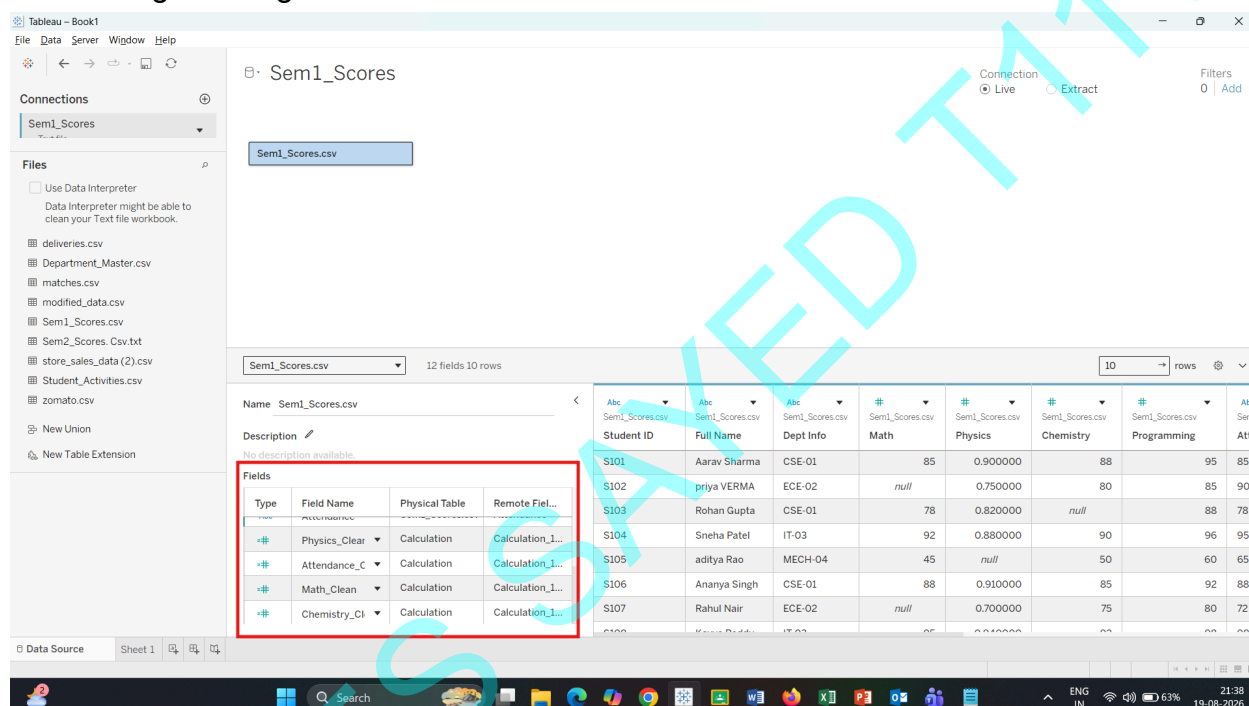
SUBJECT:- Tableau

colour coding, subject-wise side-by-side bar charts, and streamlined slice-and-dice filtering in Tableau.

Step 1: Open the Data Source Grid

1. In Tableau, navigate to the Data Source tab at the bottom-left corner.
2. Locate the four horizontal subject score columns:

- Math
- Physics
- Chemistry
- Programming



Step 2: Multi-Select and Pivot Columns

1. Click the header of the Math column.
2. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue.
3. Right-click on any of the highlighted column headers.
4. Select Pivot.

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

deliveries.csv

Department_Master.csv

matches.csv

modified_data.csv

Sem1_Scores.csv

Sem2_Scores.Csv.txt

store_sales_data (2).csv

Student_Activities.csv

zomato.csv

New Union

New Table Extension

Sem1_Scores

Sem1_Scores.csv

12 fields 10 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Text	Dept Info	Sem1_Scores.csv	Dept Info
Text	Math	Sem1_Scores.csv	Math
Text	Physics	Sem1_Scores.csv	Physics
Text	Chemistry	Sem1_Scores.csv	Chemistry
Text	Programming	Sem1_Scores.csv	Programming

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Score
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S102	priya VERMA	ECE-02	90	0.750000	80	85	90 %
S103	Rohan Gupta	CSE-01	78	0.820000	90	88	78 %
S104	Sneha Patel	IT-03	92	0.880000	90	96	95 %
S105	aditya Rao	MECH-04	45	0.700000	50	60	65 %
S106	Ananya Singh	CSE-01	88	0.910000	85	92	88 %
S107	Rahul Nair	ECE-02	90	0.700000	75	80	72 %

Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:

1. Double-click the header named Pivot Field Names and type: Subject
2. Double-click the header named Pivot Field Values and type: Score

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

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matches.csv

modified_data.csv

Sem1_Scores.csv

Sem2_Scores.Csv.txt

store_sales_data (2).csv

Student_Activities.csv

zomato.csv

New Union

New Table Extension

Sem1_Scores

Sem1_Scores.csv

10 fields 40 rows

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Field
Text	Dept Info	Sem1_Scores.csv	Dept Info
Text	Attendance	Sem1_Scores.csv	Attendance
Text	Subject	Pivot	Pivot Field Name
Text	Score	Pivot	Pivot Field Value
Text	Physics_Cle	Calculation	Calculation_1

Student ID	Full Name	Dept Info	Attendance	Subject	Pivot Score	Calculation
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	n
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	n
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	n
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	n
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	n
S102	priya VERMA	ECE-02	90 %	Math	0.7500	n
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	n

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Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure:

1. Click the Abc icon directly above the Score column header.
2. Select Number (decimal) from the dropdown list.
3. Alternatively, create a calculated field inside Sheet 1 named Clean_Score:

Code snippet

```
IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)
```

C: Perform Union of Multiple Files

Objective: Stack Sem1_Scores.csv and Sem2_Scores.csv vertically into a single unified table.

In academic reporting, performance data is recorded semester by semester in distinct files with identical schemas. Performing a Union consolidates multiple periodic files into a single long dataset, allowing you to track semester-over-semester score trends, compare Semester 1 versus Semester 2 performance for each student, and compute overall year-long averages.

Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.

Step 2: Create a Multi-File Union

1. Look at the left sidebar under the Files / Connections pane.
2. Drag the New Union option onto the empty canvas area.
3. In the pop-up Union dialog box:
 - Drag Sem1_Scores.csv from the left panel into the box.
 - Drag Sem2_Scores.csv from the left panel into the same box.
 - Verify that both filenames are listed in the Union box.
4. Click Apply, then click OK.
5. The canvas table icon will now display with a plus symbol (+), representing the consolidated union.

Step 3: Verify the Merged Data Grid

1. Review the data grid at the bottom of the screen.
2. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).
3. Scroll to the far right of the grid to verify the generated system column:

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- Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.

The screenshot shows the Tableau interface with a Union of two tables. The 'Table Name' field is highlighted with a red box, showing 'Sem1_Scores.csv' for the first 10 rows and 'Sem2_Scores.csv' for the next 10 rows. The data table below shows the results of the Union.

Union	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name
CSE-01	88	0.910000	85	92	88 %	Sem1_Scores.csv	
ECE-02	82	0.700000	75	80	72 %	Sem1_Scores.csv	
IT-03	95	0.940000	92	98	98 %	Sem1_Scores.csv	
CSE-01	67	0.600000	58	72	80 %	Sem1_Scores.csv	
ECE-02	82	0.850000	80	89	91 %	Sem1_Scores.csv	
CSE-01	88	0.920000	90	98	88 %	Sem2_Scores.c...	
ECE-02	79	0.780000	82	88	92 %	Sem2_Scores.c...	
CSE-01	82	0.850000	80	90	80 %	Sem2_Scores.c...	

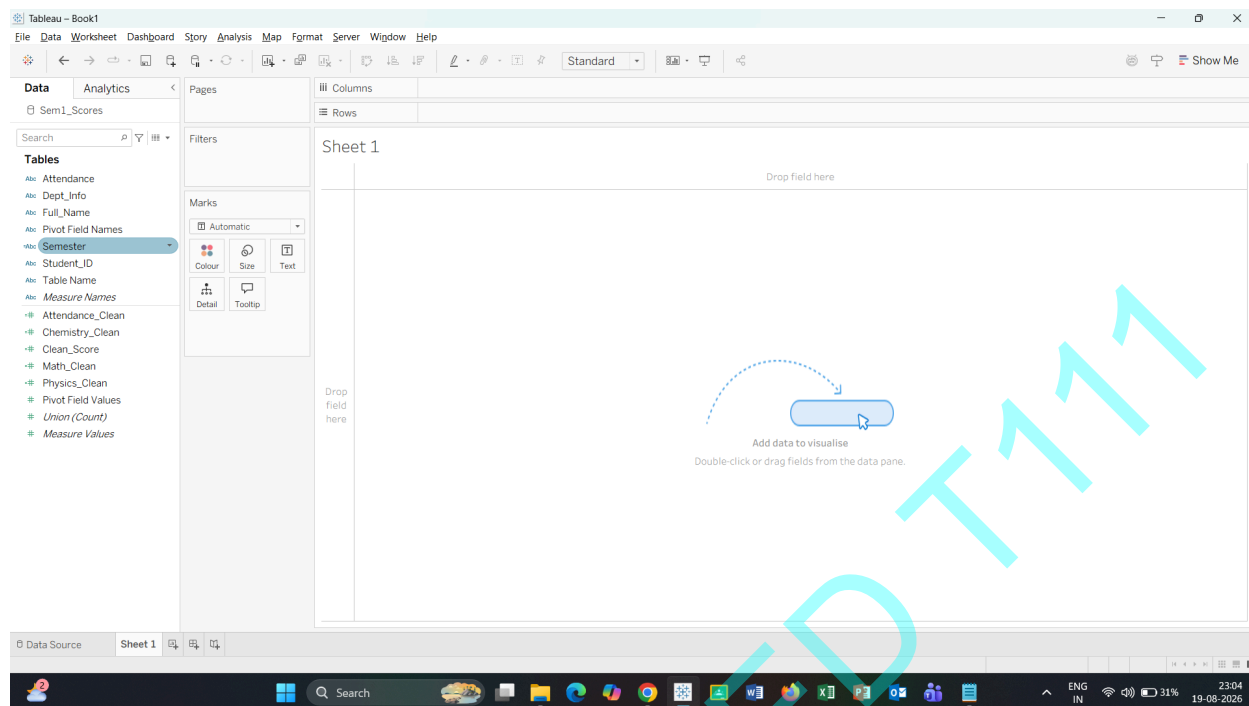
Step 4: Create a Clean Semester Identifier Dimension

- Click on Sheet 1 at the bottom-left.
- In the Data Pane (left sidebar), right-click in an empty space select Create Calculated Field....
- Set the Field Name to: Semester
- Enter the following formula: Code snippet

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```
- Click OK.

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D: Execute Cross-Database Joins

Objective: Merge your unioned student scores dataset with two independent tables: Department_Master.csv (using Dept_Info = Dept_Code) and Student_Activities.csv (using Student_ID).

Student academic scores rarely include institutional metadata (like Department Names or Targets) or extracurricular records. A Physical Join bridges separate tabular datasets horizontally across common primary/foreign keys, creating an enriched single record for each student.

Step 1: Stay on the Top Canvas

- Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv.
2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.
3. A flexible line (relationship / join noodle) will connect the two boxes.
4. If the relationship dialog opens (or click the relationship line):
 - Left Table Field (Union): Select Dept_Info
 - Right Table Field (Department_Master): Select Dept_Code
 - Operator: =

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

deliveries.csv

Department_Master.csv

matches.csv

modified_data.csv

Sem1_Scores.csv

Sem2_Scores.csv

store_sales_data (2).csv

Student_Activities.csv

zomato.csv

New Union

New Table Extension

Union

Department_Master.csv

How do relationships differ from joins? Learn more

Union Operator Department_Master...

As: Dept Info = As: Dept Code

Add more fields

Performance Options

Remove Relationship

Dept Code	Department Name	HOD Name	Dept Target Average	Total Dept Capacity
CSE-01	Computer Science	Dr. Raman	85	120
ECE-02	Electronics & Comm	Dr. Iyer	78	90
IT-03	Information Tech	Dr. Sen	82	60
MECH-04	Mechanical Engg	Dr. Sharma	70	60

Data Source Sheet 1

23:12 19-08-2026

Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.
2. Drop it to the right of the Union box.

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

deliveries.csv

Department_Master.csv

matches.csv

modified_data.csv

Sem1_Scores.csv

Sem2_Scores.csv

store_sales_data (2).csv

Student_Activities.csv

zomato.csv

New Union

New Table Extension

Union

Department_Master.csv

Student_Activities.csv

How do relationships differ from joins? Learn more

Union Operator Student_Activities...

As: Student_ID = As: Student ID (Stu...

Add more fields

Performance Options

Remove Relationship

Student ID (Student Activ...	Club Membership	Projects Completed	Certifications
S1001	Coding Club	3	2
S1002	Robotics Club	2	1
S1003	Coding Club	2	1
S1004	Debate Society	4	3
S1005	Sports Club	1	0
S1006	Coding Club	3	2
S1007	Robotics Club	2	1

Data Source Sheet 1

23:17 19-08-2026

3. In the relationship dialog:

- Left Table Field (Union): Select Student_ID
- Right Table Field (Student_Activities): Select Student_ID

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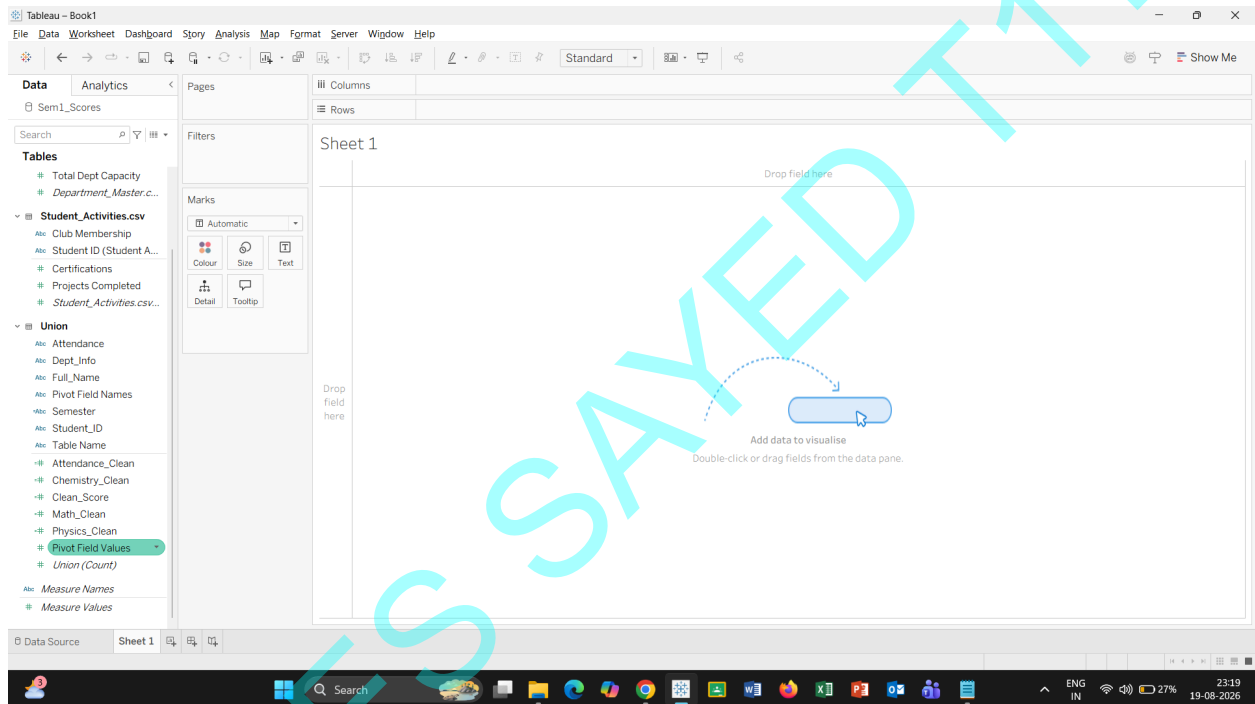
- Operator: =

Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.

2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:

- Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
- Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
- Student_Activities: Club_Membership, Projects_Completed, Certifications



E: Restructure Data for Better Visualization

Objective: Create standardized metric fields, calculated performance tiers, and formatted indicators so that raw marks transform into an intuitive student self-performance dashboard.

Raw numerical scores alone do not tell a student whether they are meeting performance criteria or lagging behind. By restructuring the data into calculated total marks, percentage scores, performance bands (Distinction, First Class, Needs Support), and attendance flags, we prepare the dataset for high-impact visual scorecards, color-coded heatmaps, and target-vs-actual comparison charts.

Step 1: Calculate Total Marks and Scaled Percentage

1. Navigate to Sheet 1 in Tableau.

2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....

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3. Create the student's total marks field:

- Field Name: Total_Score
- Tableau Formula: Code snippet
 $[Math_Clean] + [Physics_Clean] + [Chemistry_Clean]$
- Click OK.

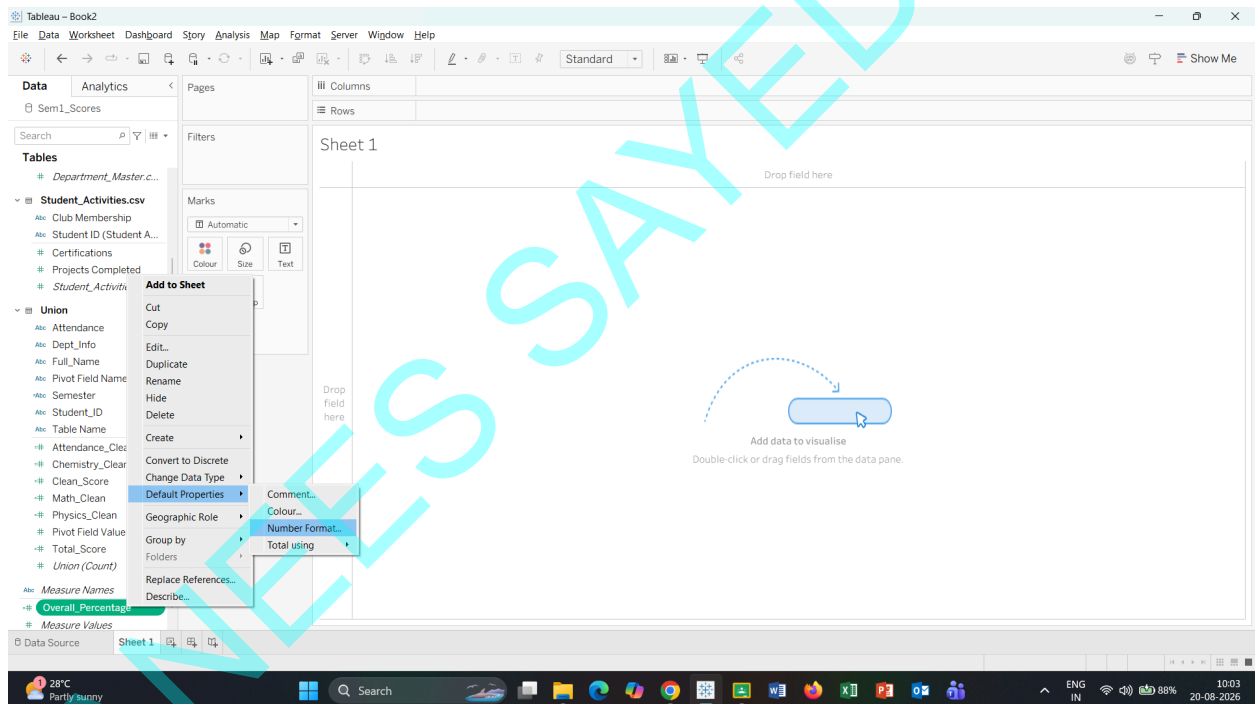
4. Create the scaled overall percentage field:

- Right-click in the Data Pane --> select Create Calculated Field....
- Field Name: Overall_Percentage
- Tableau Formula: Code snippet
 $[Total_Score] / 400.0$

Click OK.

5. Set the default percentage display formatting:

- In the Data Pane, right-click Overall_Percentage --> Default Properties --> Number Format....

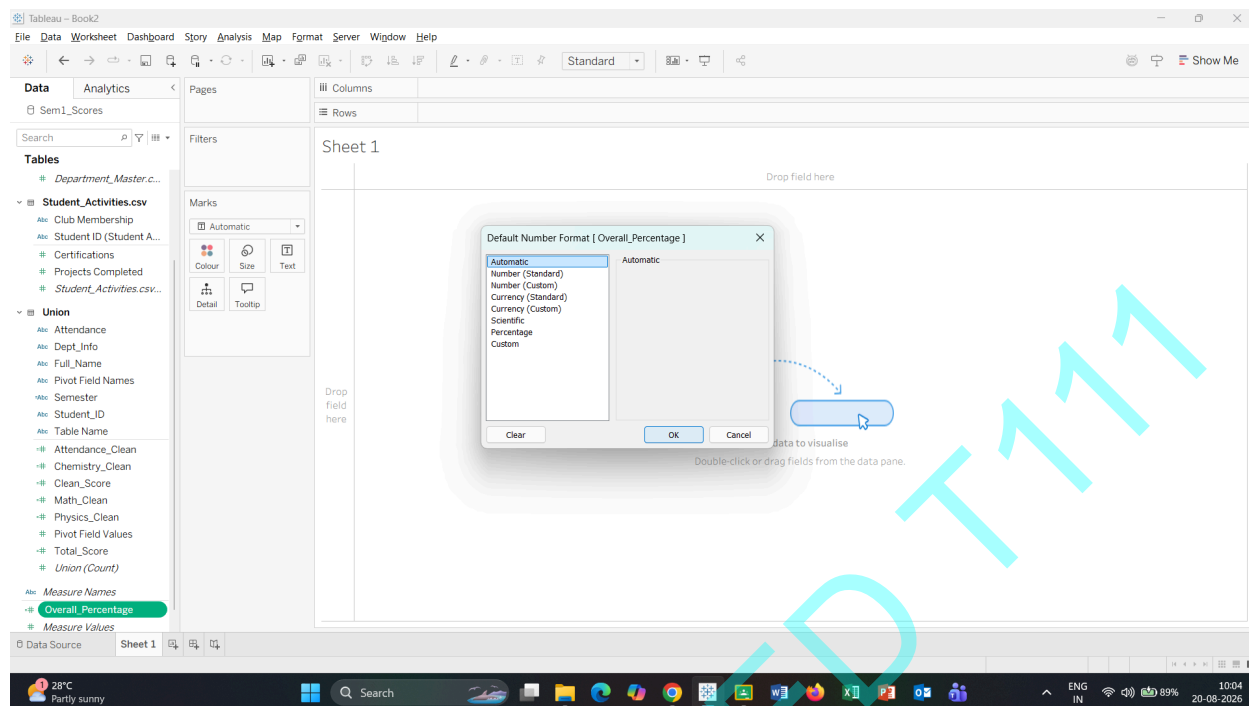


o Select Percentage and set decimal places to 1 (e.g., 85.5%).

o Click OK.

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Step 2: Create Categorical Performance Bands (Grading Tiers)

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create the academic grading tier field:

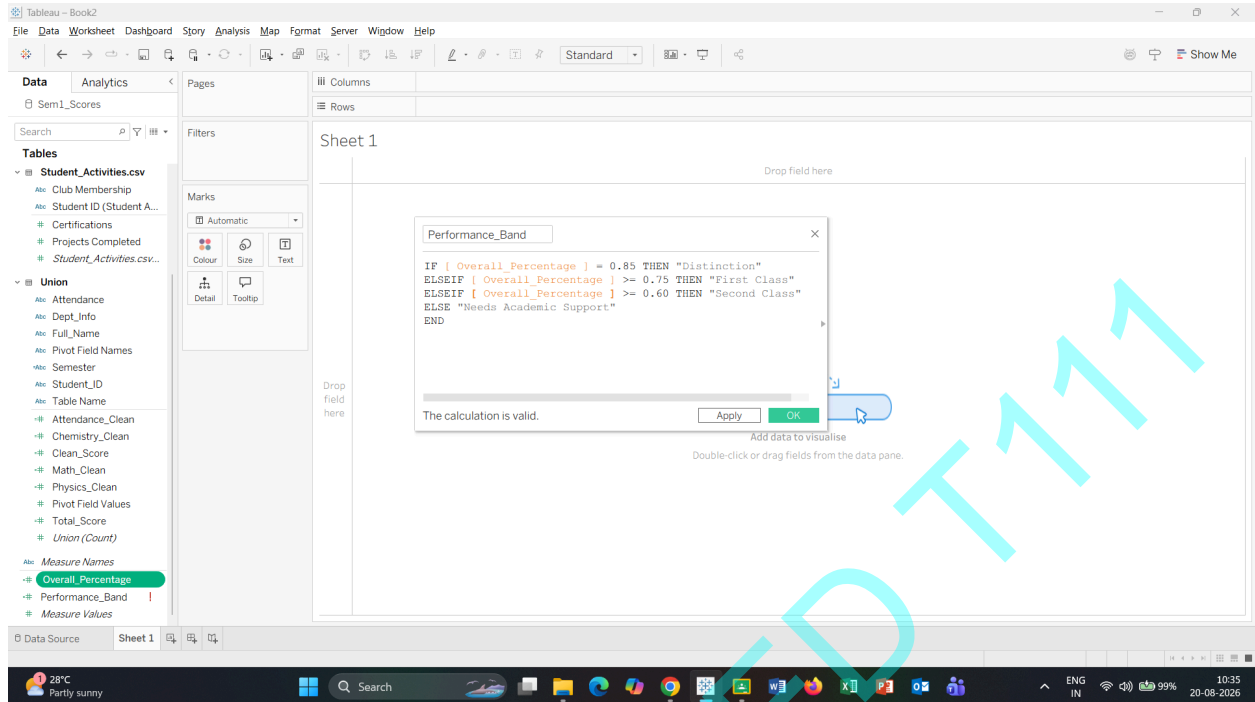
- Field Name: Performance_Band
- Tableau Formula:
- Code snippet

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END
```

Click OK.

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Step 3: Create an Attendance Health Flag

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create the attendance compliance indicator:

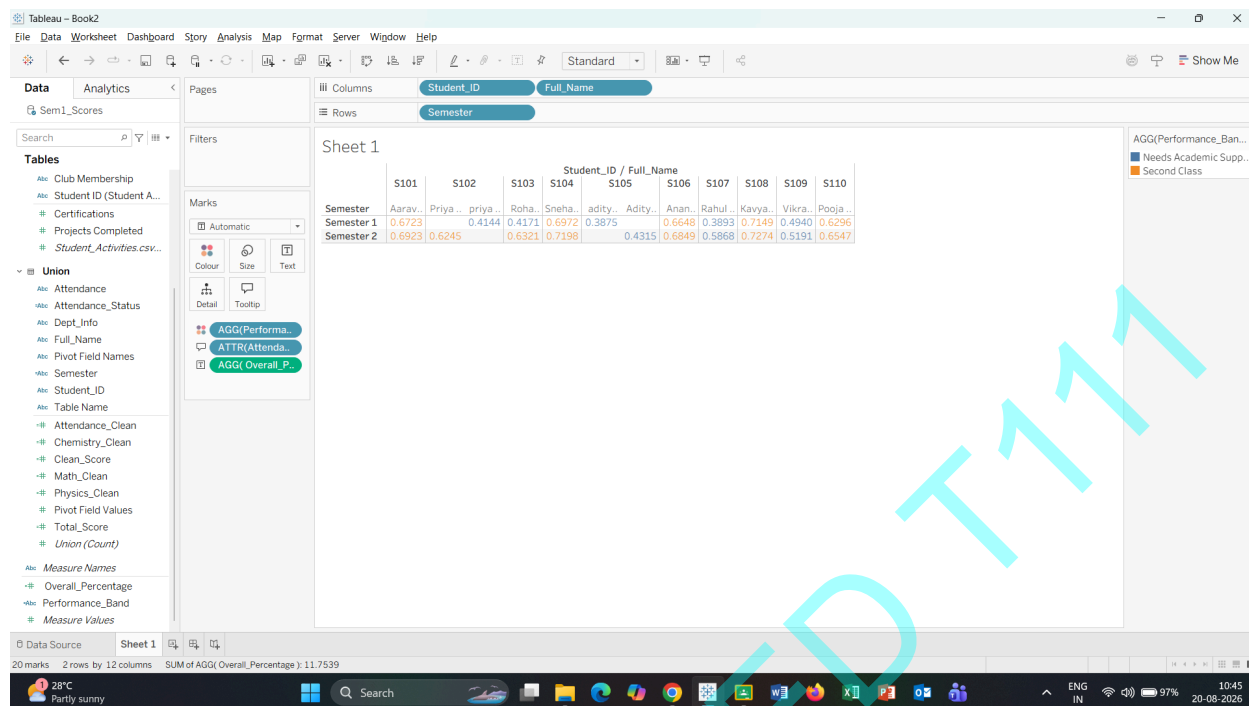
- Field Name: Attendance_Status
 - Tableau Formula:
IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"
ELSE "Shortage Warning (<75%)"
END
 - Code snippet
- Click OK.

Step 4: Build the Student Performance Scorecard View

1. Drag Student_ID and Full_Name onto the Rows shelf.
2. Drag Semester onto the Columns shelf.
3. Drag Overall_Percentage onto the Text tile on the Marks card.
4. Drag Performance_Band onto the Color tile on the Marks card.
5. Drag Attendance_Status onto the Tooltip (or Detail) tile on the Marks card.

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F: Handle Different Levels of Detail in Datasets

Objective: Compute metrics across different structural granularities (e.g., student-level score vs. department-wide average vs. overall college benchmark) on the same visualization row using Level of Detail (LOD) expressions.

A raw student score (e.g., 85%) only reflects individual performance. To understand whether that student is outperforming their peers, we must aggregate at the Department Level or College Level without collapsing the individual student rows. Tableau's {FIXED} LOD expressions allow calculations at a broader granularity independent of the dimensions placed on the visualization shelves.

Step 1: Compute Department-Level Benchmark using {FIXED} LOD

1. In Tableau, navigate to Sheet 1 (or create a new worksheet named Sheet 2).
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....

3. Create the department-wide benchmark field:

o Field Name: Dept_Average_Percentage_LOD

o Tableau Formula:

Code snippet

```
{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }
```

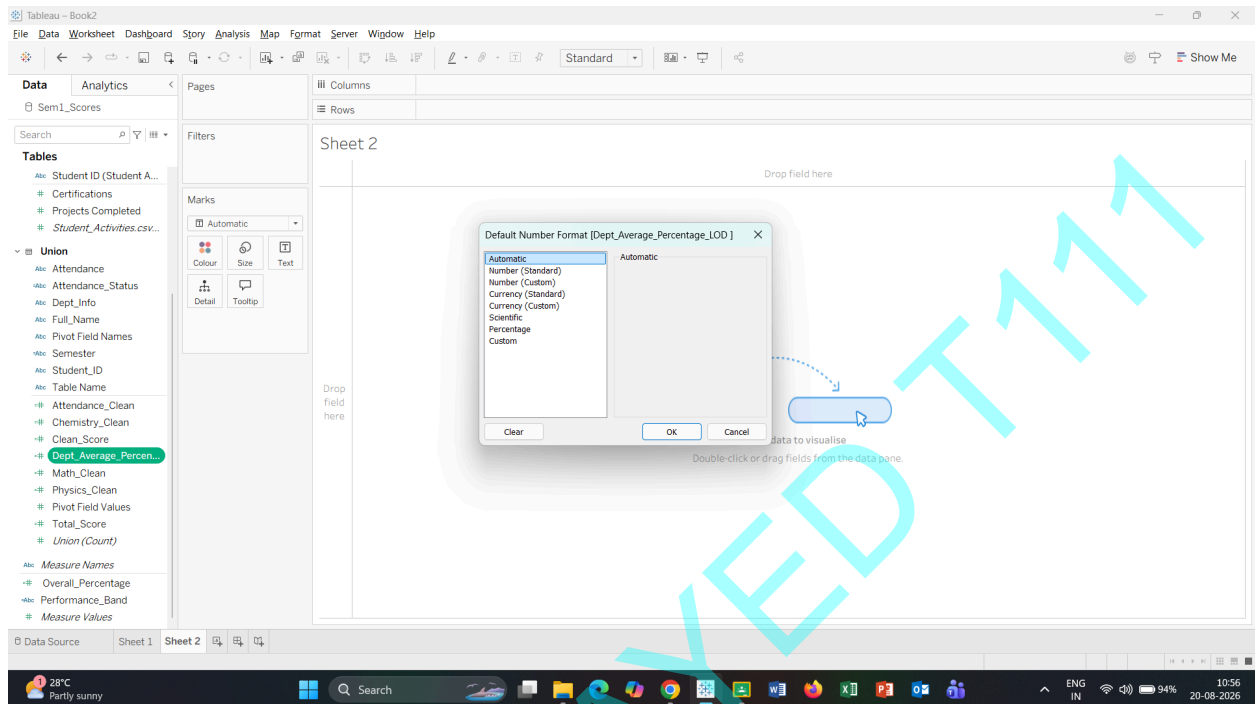
Click OK.

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SUBJECT:- Tableau

4. Set default formatting:

- Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.



Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

1. Right-click in the Data Pane --> select Create Calculated Field....

2. Create the cumulative student benchmark across all semesters:

- Field Name: Student_Overall_Career_Average
- Tableau Formula: Code snippet
`{ FIXED [Student_ID] : AVG([Overall_Percentage]) }`

Click OK.

3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).

Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

1. Right-click in the Data Pane --> select Create Calculated Field....

2. Create the relative difference metric:

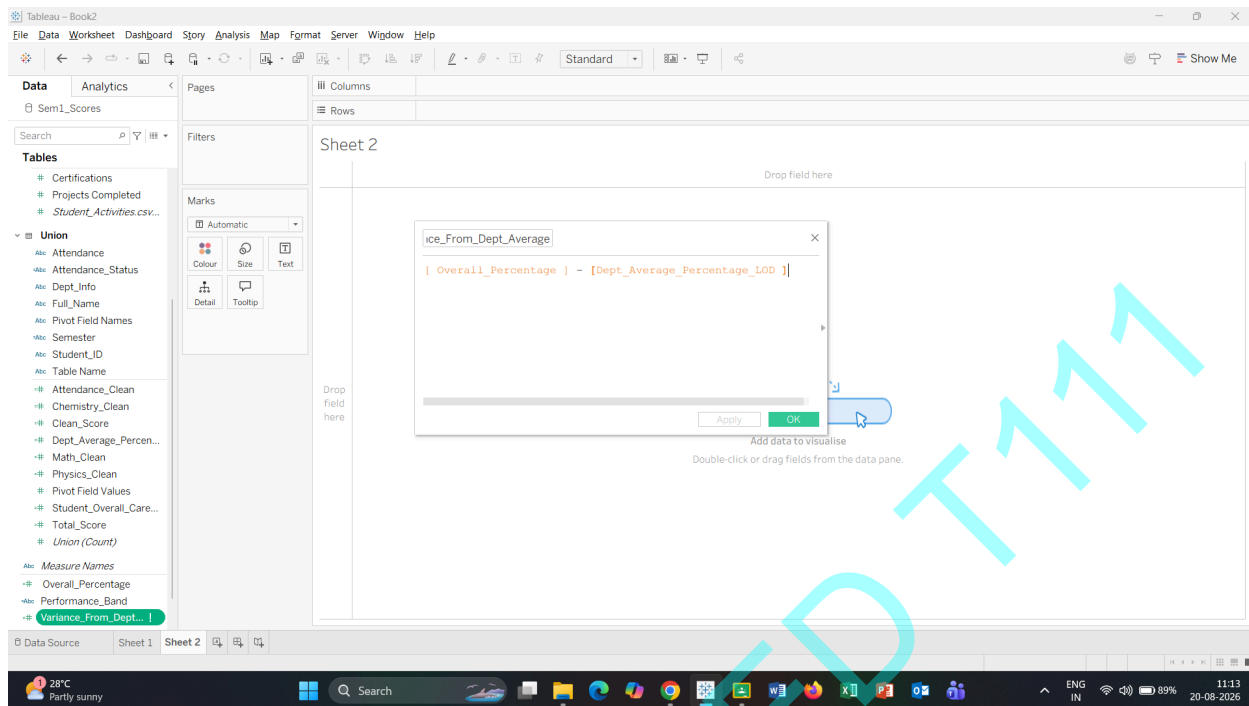
- Field Name: Variance_From_Dept_Average
- Tableau Formula: `[Overall_Percentage] - [Dept_Average_Percentage_LOD]`
- Click OK.

●

Code snippet

3. Set its Number Format to Percentage (1 decimal place).

SUBJECT:- Tableau



Step 4: Create a Benchmark Indicator Dimension

1. Right-click in the Data Pane -->select Create Calculated Field....
2. Create an automated comparison label:

- Field Name: Benchmark_Status
- Tableau Formula:
Code snippet
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark"
END

Click OK.

Step 5: Build the LOD Benchmark Visualization

1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.
2. Drag Semester to the Columns shelf.
3. Drag Overall_Percentage onto Text on the Marks card.
4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.
5. Drag Benchmark_Status onto Color on the Marks card.

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Dept_Info	Student_ID	Full_Name	Semester
CSE-01	S101	Aarav Sharma	0.6723 0.6923
	S103	Rohan Gupta	0.4171 0.6321
	S106	Ananya Singh	0.6648 0.6849
	S109	Vikram Joshi	0.4940 0.5191
ECE-02	S102	Priya Verma	0.6245
		priya VERMA	0.4144
	S107	Rahul Nair	0.3893 0.5868
IT-03	S110	Pooja Iyer	0.6296 0.6547
	S104	Sneha Patel	0.6972 0.7198
MECH-04	S108	Kavya Reddy	0.7149 0.7274
	S105	Aditya Rao	0.4315
		aditya Rao	0.3875

G: Apply Advanced Data Cleaning Techniques

Objective: Parse, standardize, and extract structured components from irregular text fields (such as leading/trailing whitespace, inconsistent letter casing, and delimited composite strings).

In student records, names often contain formatting defects (e.g., ' Aarav Sharma ' or 'priya VERMA'), while operational codes like 'CSE-01' combine the academic branch and batch number in a single column. Applying text-parsing functions like TRIM(), SPLIT(), and casing standardizations ensures clean labels and enables multi-level categorization without altering the raw database files.

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

1. Navigate to Sheet 1 (or a new worksheet).
2. Right-click an empty space in the Data Pane --> select Create Calculated Field....
3. Create the trimmed and standardized student name field:

- Field Name: Full_Name_Clean

- Tableau Formula:

Code snippet

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") - 1)) + " " +
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1, 1)) +
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2,
LEN(TRIM([Full_Name]))))
```

(Alternatively, for standard trim-only cleaning):

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SUBJECT:- Tableau

Code snippet TRIM([Full_Name])

Click OK.

Step 2: Parse First Name and Last Name using SPLIT()

1. Extract the First Name:

Right-click in the Data Pane --> select Create Calculated Field....

Field Name: First_Name

Tableau Formula:

Code snippet

TRIM(SPLIT([Full_Name], " ", 1))

Click OK.

2. Extract the Last Name:

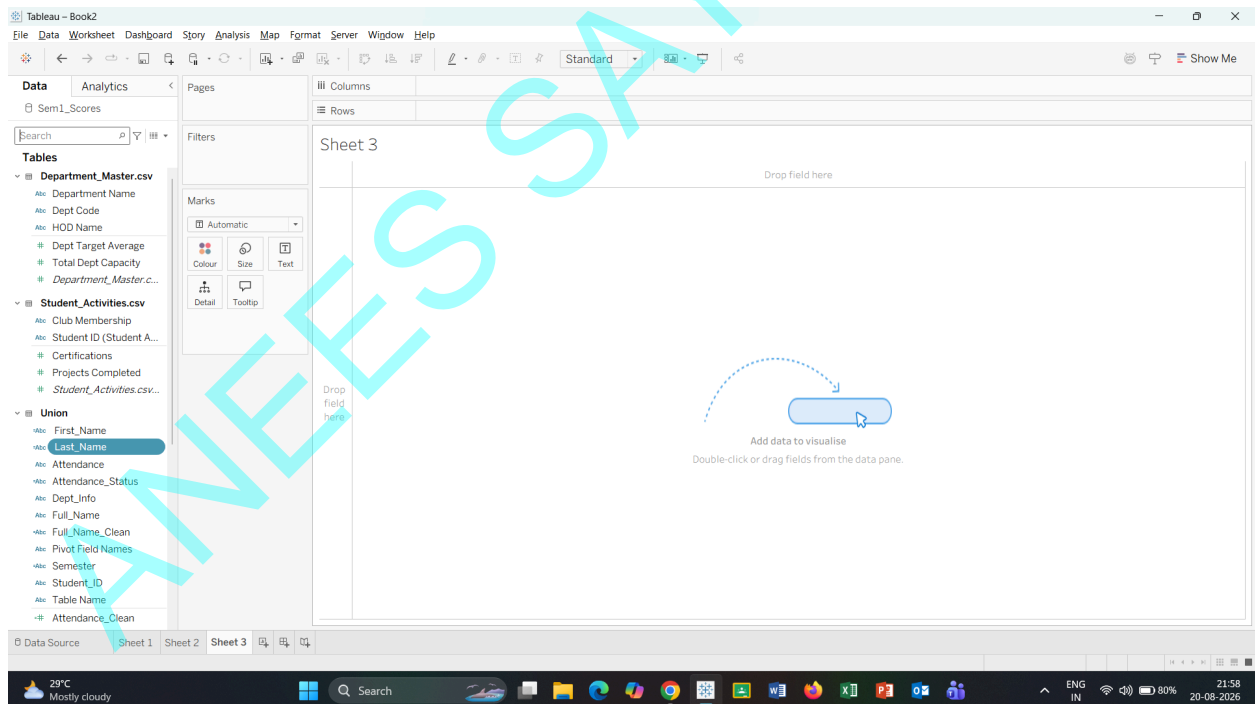
- Right-click in the Data Pane --> select Create Calculated Field....
- Field Name: Last_Name

Tableau Formula:

Code snippet

TRIM(SPLIT([Full_Name], " ", -1))

Click OK.



SHETH MVLU COLLEGE

SUBJECT:- Tableau

Step 3: Split Composite Department Codes (Dept_Info)

The column Dept_Info contains combined strings like CSE-01, ECE-02, and MECH-04. Separate them into distinct dimensional attributes:

1. Extract the Department Acronym:

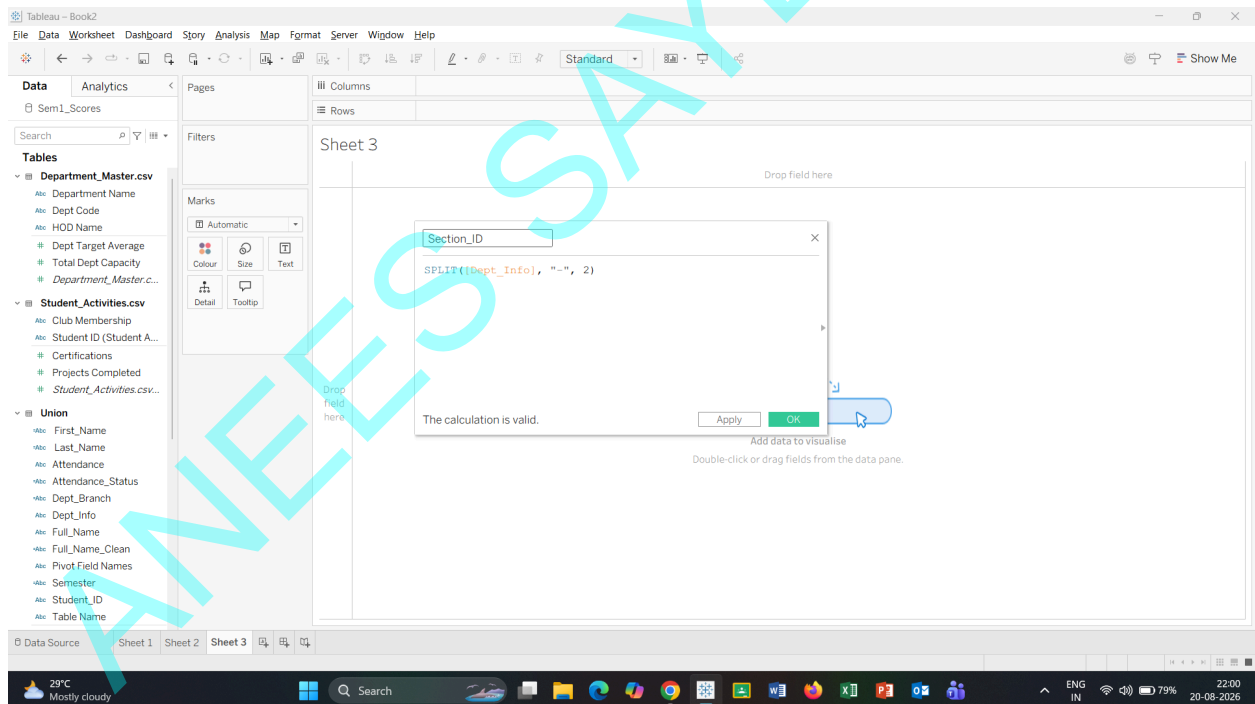
- Right-click in the Data Pane --> select Create Calculated Field....
- Field Name: Dept_Branch
- Tableau Formula:
Code snippet
`SPLIT([Dept_Info], "-", 1)`

Click OK.

2. Extract the Batch / Section ID:

- Right-click in the Data Pane --> select Create Calculated Field....
- Field Name: Section_ID
- Tableau Formula:
Code snippet
`SPLIT([Dept_Info], "-", 2)`

Click OK.



Step 4: Create a Clean Composite Student Label

Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

1. Right-click in the Data Pane --> select Create Calculated Field....

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SUBJECT:- Tableau

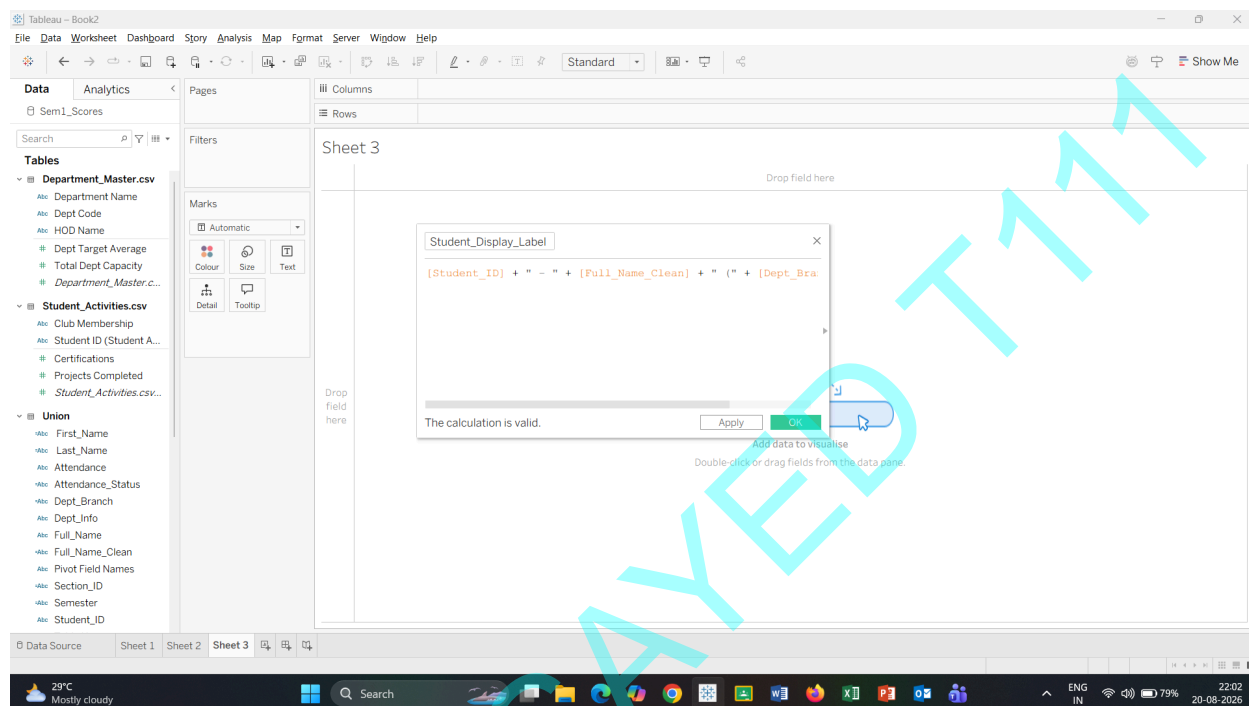
2. Field Name: Student_Display_Label

3. Tableau Formula:

Code snippet

[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"

4. Click OK.



Step 5: Verify in the Final Dashboard View

1. Place Student_Display_Label on the Rows shelf.

2. Place Semester on the Columns shelf.

3. Place Overall_Percentage on Text.

4. Place Performance_Band on Color.

5. Place Dept_Branch and Section_ID in the Filters shelf. (While Filtering You Can select any of the value you want)

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SUBJECT:- Tableau

